



REVIEW

The Burden of Delayed Diabetic Retinopathy Management and Use of Artificial Intelligence-Driven Screening Tools: A Systematic Literature Review

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ABSTRACT

Purpose: Patients with diabetic retinopathy (DR) are at risk of visual deterioration owing to systemic and financial barriers in accessing appropriate care. DR screening tools that implement artificial intelligence (AI) algorithms are gaining recognition due to their accuracy and high-throughput potential. This systematic literature review aimed to understand the economic, humanistic, and clinical burden associated with delayed DR management and the

impact of AI-based screening tools for diagnosis and treatment.

Methods: MEDLINE, Embase, and Cochrane Library databases were searched per Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines (1 January 2014 to 28 October 2024). Screening, extraction, and quality assessment were performed by two independent reviewers. Supplementary searching was conducted to evaluate visual outcomes.

Results: In total, 33 records were included. Economic evidence demonstrated that infrequent screening was initially cost-saving but decreased patient quality-adjusted life years, delayed sight-threatening DR diagnosis, and resulted in high treatment-related costs in the long term. Several studies found delayed DR treatment to adversely impact visual acuity, central subfield thickness, and time spent with vision loss. The majority of economic studies evaluating AI-based screening found its use to result in lower overall costs than conventional screening, while two noted higher costs attributable to greater screening uptake and increased specialist referrals. Most studies that modeled clinical impact found AI-based screening to reduce blindness or vision loss versus conventional screening.

Conclusions: This research underscored the considerable harms associated with delayed DR diagnosis and treatment. AI-based screening tools have the potential to become powerful instruments in supporting improved clinical

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outcomes for patients and economic benefits for healthcare systems.

Keywords: Diabetic retinopathy; Artificial intelligence; Vision loss; Screening programs; Delayed diagnosis; Systematic literature review

Key Summary Points

Delayed diagnosis and management of diabetic retinopathy (DR) contributes to otherwise avoidable vision loss and considerable healthcare and economic burden.

Artificial intelligence (AI)-driven screening tools offer a potential solution for early detection, but their real-world performance, economic value, and barriers to implementation remain unclear.

This systematic literature review evaluated the clinical, humanistic, and economic consequences of delayed DR management and assessed evidence on AI-driven screening approaches.

Across literature, delayed treatment or follow-up consistently resulted in worse visual outcomes, higher rates of macular edema or proliferative disease, and increased downstream healthcare utilization and costs. Evidence suggests that timely DR management may improve outcomes and reduce long-term costs.

AI-driven screening tools show promise, particularly in primary care or community settings where access barriers exist, but further robust economic and effectiveness data needed to optimize adoption and reimbursement pathways.

INTRODUCTION

Diabetic retinopathy (DR) is a microangiopathic disease, resulting from the chronic effects of diabetes mellitus (DM). A total of 103 million adults

worldwide are estimated to be affected by DR, which is expected to rise to 161 million individuals by 2045 [1]. In 2021, 26.43% of Americans with diabetes lived with DR, and 5.06% of those with diabetes had sight-threatening DR [2].

DR is classified as the most common retinal vascular disease, and the fifth most common global cause of moderate-to-severe visual impairment in those aged 50 years and above [3, 4]. DR typically advances from nonproliferative DR (NPDR), which involves progressive retinal vasculature permeability and capillary leakage, to proliferative DR (PDR), characterized by new blood vessel growth on the optic disc and retina, and vitreous or preretinal hemorrhage [3].

Visual deterioration is preventable through the early detection and management of DR. Early treatment can prevent or delay blindness from DR in up to 90% of patients with diabetes [5]. Timely screening of patients with diabetes is important even in the absence of symptoms, since retinal damage and microvasculature changes can progress irreversibly during this early phase [6]. Yet, many patients experience barriers in accessing timely DR screening and care. These include system-related barriers, such as a shortage of eye care professionals and screening services in certain geographic locations [7, 8], and patient-related barriers, such as knowledge gaps, lack of transportation, and financial limitations, which predominate in lower socioeconomic status groups [9, 10].

DR screening tools that augment artificial intelligence (AI) algorithms are gaining recognition for clinical use. AI-based screening tools typically use deep learning algorithms that can identify signs of DR with high sensitivity and specificity from retinal images. Various AI-based tools for DR screening have shown high accuracy levels [11]. Additionally, the low operational cost of AI-based tools has the potential to increase screening affordability and overall access [12, 13].

This systematic literature review (SLR) aimed to define the economic, humanistic, and clinical burden associated with delayed DR management and highlight the impact of AI tools for DR diagnosis and treatment.

METHODS

Systematic Review

The SLR was conducted according to Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines [14, 15] (see Fig. 1 for the PRISMA flow diagram). The review was based on previously conducted studies and did not contain any new studies with human participants or animals performed by any of the authors; therefore, ethical approval was not required.

Search Strategies

The search strategy (Supplementary Table S1) was developed by a medical information specialist and peer-reviewed using the Peer Review of Electronic Search Strategies (PRESS) checklist [16]. The database searches aimed to identify published records on: (1) the economic burden and quality of life or humanistic burden associated with delayed DR screening and (2) the impact of AI-based tools for DR screening and diagnosis. Humanistic burden was defined as the impact of DR on patient or caregiver quality of life or well-being. Searches were performed in MEDLINE, Embase, and the Cochrane Library for citations published between 1 January 2014 and 28 October 2024.

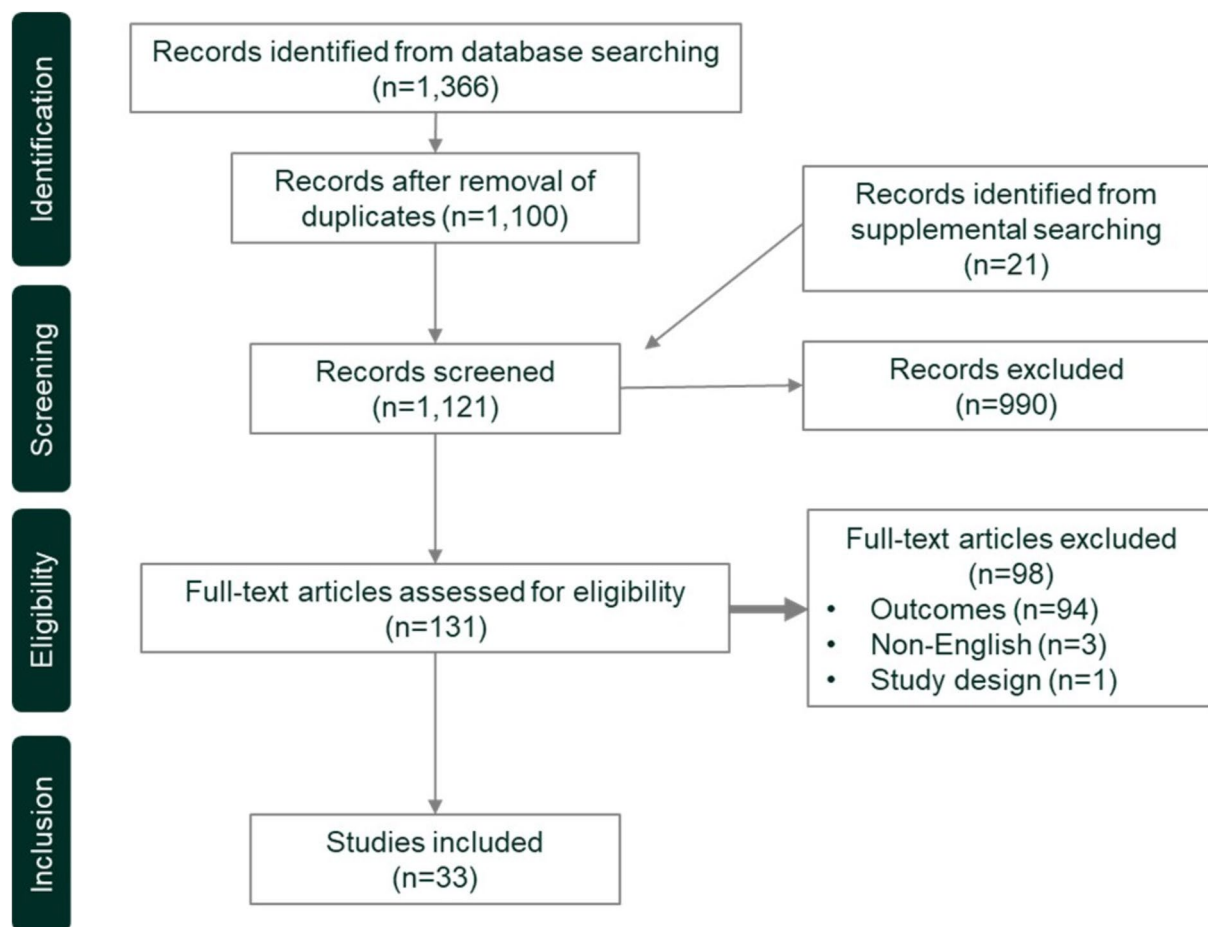


Fig. 1 PRISMA flow diagram

Supplementary searches were conducted to understand the impact of delayed DR management on visual outcomes. The same keywords and search terms were used to define the population, interventions, and outcomes of interest, with additional consideration of blindness and visual impairment by screeners when assessing study relevance.

Eligibility Criteria

Studies were selected according to predefined population, intervention/comparator, outcomes, study design (PICOS) criteria (Supplementary Table S2).

Study Selection, Data Extraction, and Study Quality Assessment

Citations were screened by two independent reviewers, who resolved conflicts through discussion. A third reviewer adjudicated unresolved conflicts. Extraction was performed similarly, where an independent reviewer extracted relevant details, and a second independent reviewer reviewed for accuracy. All costs were inflated to 2025 values and converted to 2025 United States dollar (USD) to ensure consistency and comparability across studies in the SLR. For studies that did not report cost year, costs were inflated to 2025 values from the year of publication (Table 1). Quality assessment was conducted using the Downs and Black checklist [17] for observational studies and the Drummond and Jefferson checklist [18] for economic modeling studies.

RESULTS

A total of 1100 unique articles were identified from the search, and 21 studies were identified from the supplementary search. After screening, a total of 33 records were deemed eligible for inclusion (Fig. 1). Study details are listed in Table 2.

Burden of Delayed DR Diagnosis and Treatment

Economic Burden

Seven studies discussed the economic outcomes associated with the timing and frequency of DR screening in patients with diabetes. The Spanish cost-effective analysis (CEA) compared annual screening for DR with screening every 2.5 years. Screening every 2.5 years was associated with lower incremental costs (in 2025 USD) for patients with any DR (\$1232), sight-threatening DR (\$3904), and diabetic macular edema (DME; \$4661) but also lower incremental quality-adjusted life years (QALYs) of 0.77, 0.60, and 0.44, respectively [19]. The Taiwanese propensity score-matched longitudinal 17-year study found that high-frequency screening (>once per year) and intermediate-frequency screening (once every 1–2 years) were associated with higher average DR screening costs per patient (\$31 and \$59, 2025 USD) than low-frequency screening (less than every 2 years; \$14, 2025 USD), but resulted in substantially lower treatment-related costs ($p < 0.001$), indicating that upfront costs for screening were mitigated by long-term cost savings on treatment [20]. Finally, the English retrospective cohort study of participants in the Norfolk Diabetic Retinopathy Screening Programme from 2006 to 2017 found that as patient age increased, there was a decline in the incidence of DR treatment and an increase in screening costs per person treated [21].

Four studies (from the USA, the UK, the Netherlands, and Scotland) discussed the economic outcomes of individualized screening frequencies based on disease severity and personal risk factors (i.e., screening patients at a higher risk for sight-threatening DR or PDR more often than patients who are at a lower risk) compared with nonstandardized, indiscriminate frequencies for screening across all patients [22–25]. All four studies found risk-stratified screening approaches to result in cost savings related to screening; however, the Dutch and Scottish studies found risk-stratified screening frequencies to increase the number of delayed sight-threatening

Table 1 Inflation and conversion of costs used in this SLR

Study	Currency and year	Cost in study	Cost inflated to 2025	Cost converted to 2025 USD ^a
Chakravarthy (2024)	CAD, 2024 ^b	\$197,600,000.00	\$201,045,080.95 ^c	\$145,255,070.99
		\$84,000,000.00	\$85,464,508.09 ^c	\$61,748,107.10
Chawla (2023)	USD, 2021	\$301.11	\$369.14 ^a	\$369.14
Chung (2024)	New Taiwanese dollar, 2024 ^b	\$930.00	\$949.53 ^d	\$31.33
		\$1751.00	\$1787.77 ^d	\$59.00
		\$427.00	\$435.97 ^d	\$14.39
Fuller (2022)	USD, 2019	\$485.92	\$619.28 ^a	\$619.28
Hu (2024)	AUD, 2022	\$595,800,000.00	\$649,038,216.81 ^c	\$418,434,938.38
Karabeg (2024)	USD, 2022	\$143.00	\$163.17 ^a	\$163.17
Romero-Aroca (2016)	Euro, 2016 ^d	€865.57	€1109.17 ^f	\$1232.40
		€2742.72	€3513.22 ^f	\$3903.54
		€3275.09	€4194.61 ^f	\$4660.63
Srisubut (2023)	USD, 2021	\$2.60	\$3.06 ^a	\$3.06
		\$66.70	\$78.38 ^a	\$78.38
Xie (2020)	USD, 2015	\$15.00	\$20.59 ^a	\$20.59
		\$489,000.00	\$671,219.75 ^a	\$671,219.75

AUD Australian dollar, *CAD* Canadian dollar, *USD* United States dollar

^aThe US Bureau of Labor and Statistics Consumer Price Index (CPI) Inflation Calculator (https://www.bls.gov/data/inflation_calculator.htm) was used to convert costs to 2025 USD

^bCost year was not reported in the publication. Cost year was assumed to be the same as the publication year

^cThe Bank of Canada Inflation Calculator (<https://www.bankofcanada.ca/rates/related/inflation-calculator/>) was used to inflate costs to 2025 CAD

^dAn Inflation calculator tool was used to inflate costs to 2025 New Taiwanese dollar: (<https://www.worlddata.info/asia/taiwan/inflation-rates.php>)

^eThe Reserve Bank of Australia Inflation Calculator (<https://www.rba.gov.au/calculator/annualDecimal.html>) was used to inflate costs to 2025 AUD

^fAn Inflation calculator tool was used to inflate costs to 2025 euros: (<https://www.inflationtool.com/euro/2015-to-present-value?amount=2&year2=2025&frequency=yearly>)

retinopathy diagnoses by 11% and incidence of severe vision loss by 31–46 per 100,000, respectively, compared with annual screening for all patients.

Humanistic and Clinical Burden

No studies exclusively discussed the humanistic burden (defined as the impact of DR on patient

Table 2 Study details for publications included in the review

Study source and region	Study design	Objectives and comparator groups used to inform the study	Outcomes used to inform the SLR
<i>Burden of delayed DR diagnosis and treatment</i>			
Ahearn (2023) [31], USA	Retrospective case–control study	Investigation of the impact of delayed retinal clinical care during the coronavirus disease 2019 (COVID-19) pandemic on the severity of PDR	Vitrectomy requirement
Broadbent (2021) [25], UK	RCT and CEA	Evaluation of the safety and cost-effectiveness of individualized, variable-interval, risk-based population screening compared with usual care	Costs
Brodie (2024) [21], UK	Retrospective cohort study	Evaluation of treatment probability according to age at first screening episode	Costs
Bulut (2024) [32], Turkey	Retrospective study	Assessment of the clinical impact of delayed anti-VEGF injections among treatment-naïve patients with DME and nonproliferative DR during the COVID-19 lockdown	Visual acuity
Chung (2024) [20], Taiwan	17-year longitudinal, population-based, PSM study	Evaluation of the association between DR screening frequency and the development of DR and corresponding medical expenses	Costs
DCCT/EDIC Research Group (2017) [23], USA	Markov model	Development of an individualized DR screening schedule to efficiently detect potentially vision-threatening changes	Costs associated with individualized DR screening frequencies

Table 2 continued

Study source and region	Study design	Objectives and comparator groups used to inform the study	Outcomes used to inform the SLR
Emamipour (2020) [24], the Netherlands	Economic model	Examination of the cost-effectiveness of three different screening strategies for DR: personalized adaptive model, annual screening, and Dutch guidelines	Costs associated with individualized DR screening frequencies
Fassbender (2016) [34], USA	Retrospective, single-center study	Comparison of immediate versus delayed vitrectomy for the management of vitreous hemorrhage due to PDR	Visual acuity, time spent with vision loss
Gomel (2022) [36], Israel	Retrospective, multicenter study	Evaluation of the impact of unplanned treatment gaps during the COVID-19 pandemic lockdowns on visual acuity in previously treated patients with DME	Visual acuity
Hentati (2021) [37], USA	Retrospective, single-center study	Comparison of outcomes between patients with DME and BVA of 20/25 or better who received early, delayed, or no anti-VEGF treatment	ETDRS letters lost
Luu (2021) [29], USA	Retrospective, single-center study	Real-world evaluation of the management and long-term outcomes of patients with DME and good initial visual acuity	Visual acuity
Naravane (2021) [33], USA	Retrospective, multicenter, case series study	Evaluation of the impact of > 2 weeks of delay in intravitreal anti-VEGF treatment during the COVID-19 pandemic on visual and anatomic outcomes in DME ± PDR	Visual acuity, CST

Table 2 continued

Study source and region	Study design	Objectives and comparator groups used to inform the study	Outcomes used to inform the SLR
Ohlhausen (2021) [28], USA	Retrospective clinical study	Investigation of the effect of delayed PRP treatment on visual outcomes in patients with PDR	Visual acuity
Romero-Aroca (2016) [19], Spain	CEA	CEA of a 1-year versus 2.5-year screening program	Costs, QALYs
Rush and Rush (2021) [35], USA	Retrospective, controlled case series	Evaluation of outcomes in delayed anti-VEGF treatment during the COVID-19 pandemic among patients with DME	Visual acuity
Scanlon (2014) [27], UK	Retrospective, observational study (English NHS DR Screening Programme)	Evaluation of the relationship between delayed DR screening after type 2 diabetes diagnosis and level of retinopathy detected	Trend relating time between diabetes diagnosis and DR screening to retinopathy level
Scanlon (2016) [26], UK	Retrospective cohort study (Scottish, Welsh & Northern Irish DR screening programs)	Assessment of the relationship between delayed DR screening after type 2 diabetes diagnosis and level of retinopathy detected	Trend relating time between diabetes diagnosis and DR screening to retinopathy level
Scotland (2016) [22], Scotland	CEA	Assessment of the cost-effectiveness of adopting risk-stratified approaches to extended screening intervals in the national DR screening program	Costs
Song (2021) [30], USA	Retrospective cohort study	Evaluation of the impact of delayed or missed intravitreal injection visits on visual outcomes in patients with DME, PDR, or both during the COVID-19 pandemic	ETDRS letters lost

Table 2 continued

Study source and region	Study design	Objectives and comparator groups used to inform the study	Outcomes used to inform the SLR
<i>AI for DR screening</i>			
Chakravarthy (2024) [47], Canada	CEA	CEA of two different approaches for scaling DR screening, with one approach involving the use of AI prediction models	Costs, rate of vision loss
Channa (2023) [51], USA	Policy model	Policy model to evaluate the differential impact of autonomous AI-based DR screening versus conventional screening performed in the clinic by an eye care provider	Vision loss incidence
Chawla (2023) [39], USA	CEA	CEA of AI-based fully automated retinal image screening (FARIS) versus referral-based screening for DR in the US healthcare system	Costs
Fuller (2022) [38], USA	CEA	CEA of AI-based Automated Retinal Image Analysis System (ARIAS) versus referral-based screening for DR in a primary care setting serving a low-income population	Costs, vision loss incidence
Harding (2023) [48], China	CEA; conference abstract	Evaluation of the long-term cost-effectiveness of AI-based screening versus ophthalmologist-led opportunistic screening in urban China	ICER, costs, blindness avoided
Hu (2024) [46], Australia	CEA	CEA of implementing AI-based DR screening systems in Australian primary care settings versus the status quo screening system in Australia	ICER, cost, blindness avoided

Table 2 continued

Study source and region	Study design	Objectives and comparator groups used to inform the study	Outcomes used to inform the SLR
Huang (2022) [42], China	CEA	CEA of AI-based DR screening versus ophthalmologist-led screening in rural China	ICER, costs
Karabeg (2024) [45], Norway	Cost-analysis study	Cost analysis comparing AI-grading or DR versus manual/ophthalmologist grading in a cohort of minority women with DM in Oslo	Costs
Li (2023) [41], China	CEA	CEA of AI-based versus ophthalmologist-led DR screening in rural China	ICER, costs
Lin (2023) [40], China	CEA	CEA of AI-based versus manual grading for community-based telemedicine DR screening in urban China	Costs
Mansouri (2023) [44], Algeria	BIM; conference abstract	BIA assessing the economic value of AI-based methods for DR screening	Costs
Gomez-Rossi (2022) [49], Brazil	CEA	Evaluation of whether existing AI algorithms are cost-effective for use as decision-support systems for detecting DR	Costs
Srisubut (2023) [50], Thailand	CUA	CEA of deep-learning, AI-based DR screening versus human grader-based DR screening in a middle-income country	Costs
Xie (2020) [43], Singapore	CMA	CMA evaluating the potential savings of two deep learning approaches (semi-automated and fully automated) compared with the current human assessment approach	Costs

Table 2 continued

AI artificial intelligence, *BIA* budget impact analysis, *BIM* budget impact model, *BCVA* best-corrected visual acuity, *BVA* best visual acuity, *CEA* cost-effectiveness analysis, *CMA* cost-minimization analysis, *CSA* cost-sensitivity analysis, *CST* central subfield thickness, *CUA* cost-utility analysis, *DM* diabetes mellitus, *DME* diabetic macular edema, *ETDRS* Early Treatment Diabetic Retinopathy Study, *ICER* incremental cost-effectiveness ratio, *nAMD* neovascular age-related macular degeneration, *PDR* proliferative diabetic retinopathy, *PRP* panretinal photocoagulation, *PSM* propensity score-matched, *QALY* quality-adjusted life year, *UK* United Kingdom, *USA* United States, *VEGF* vascular endothelial growth factor, *SLR* systematic literature review

or caregiver quality of life or well-being) associated with earlier or later timing of DR diagnosis. Although broader literature on humanistic burden is available, it was not incorporated to preserve the specific focus of the research question (i.e., screening timing). In addition, there was a lack of studies discussing the clinical burden (defined as the impact of DR on visual outcomes). As such, a supplementary search utilizing broader terminology was undertaken to fill these gaps, which uncovered several studies focused on clinical burden.

Clinical Burden of Delayed Screening for DR Two studies discussed the impact of delayed screening on visual acuity and/or disease severity. One was a large database analysis of 689,025 patients from the nationwide DR screening programs across Wales, Scotland, and Northern Ireland, and four local English programs [26], and the other was a database analysis of the English National Health Service diabetic eye screening program involving 8183 patients [27]. Both studies found that time from diabetes diagnosis to first DR screening was significantly associated with disease severity ($p<0.005$ in both studies).

Clinical Burden of Delayed Treatment of DR Ten retrospective observational studies discussed the impact of delayed treatment or care on visual acuity, vision loss, or other complications in patients with DR, PDR, or DME. Three US retrospective studies found that delayed treatment significantly impacted visual outcomes. One study of 259 patients with PDR who were treated with pan retinal photocoagulation (PRP) found visual acuity to be significantly reduced in the group who received PRP $\geq$ 31 days following PDR diagnosis compared with the group that received same-day treatment at both 12 months ($p<0.001$) and at 24 months ($p=0.03$) after treatment [28]. The other study of 100 patients with treatment-naïve DME and best-corrected visual acuity (BCVA) of 20/25 or better found an independent association between timing of treatment and final visual acuity ($p=0.017$), with each 1-week delay in initiating therapy associated with 0.014 logMAR (95% confidence interval [CI] 0.003–0.024) unit

worsening in final vision over a median follow-up time of 3 years [29]. A third US retrospective study of 150 patients with DME, PDR, or both found that intravitreal injection treatment delays due to missed patient visits during the COVID-19 pandemic resulted in visual acuity loss (-3.48 ± 1.95 letters), whereas patients who completed visits as scheduled retained stable vision (2.71 ± 1.75 ; $p=0.0203$) [30].

A fourth US retrospective study found that greater appointment delays during the COVID-19 pandemic among patients with PDR ($n=739$) were associated with a requirement for vitrectomy owing to proliferative complications ($p<0.001$) [31].

Five studies found no significant difference in visual acuity between groups with and without treatment delay [32–36], but two studies found treatment delays to be associated with a significantly larger central subfield thickness (CST) [33] and significantly more time spent with vision loss [34]. One US study unexpectedly found worse initial visual outcomes among patients in whom treatment was not delayed, but this did not translate into long-term or clinically meaningful vision loss [37]. Key outcomes from these studies are presented in Table 3.

AI for Diabetic Retinopathy Screening

Economic Studies of AI Use in DR Screening

A total of 13 economic modeling studies compared screening strategies that implement AI to conventional strategies. Details of the study design, population, comparators, inputs, and key findings are highlighted in Supplementary Table S3. Ten studies found that AI-driven strategies for screening resulted in lower costs than conventional screening, with savings ranging from 1% up to 63%. Both US studies that compared AI-based approaches for DR screening with conventional DR screening methods from the healthcare payer perspective studies found AI-based screening to result in similar quality-adjusted life year (QALY) gains to conventional screening, with cost savings of 18.8% (incremental cost: $-\$369$; 2025 USD) and 23.3%

(incremental cost: $-\$619$; 2025 USD), respectively, over 5 years [38, 39].

AI-based strategies for screening were also found to result in cost savings compared with ophthalmologist-led screening or manual grading across all three studies from China [40–42]. Savings ranged from 2.5% to 28.5%; AI-based screening resulted in dominant incremental cost-effectiveness ratios (ICERs) across two studies [41, 42] from the health system perspective, with the third study showing a QALY difference of 0.005 between AI-based screening and manual grading from the societal perspective [40]. In Singapore, Xie et al. estimated semi-automated screening to result in savings (in 2025 USD) of $\$20.59$ per patient compared with human-only assessment, resulting in savings of $\sim\$671,000$ to the Singapore healthcare system, equivalent to $\sim 20\%$ annual savings [43].

The budget impact analysis by Mansouri et al. estimated cost savings for screening with AI-based methods to range from 6% to 63% compared with conventional screening in Algeria [44]. The study by Karabeg et al. on minority women with DM in the Netherlands found AI-based screening to result in cost savings of $\$163.17$ per patient (2025 USD), amounting to savings of 52% compared with manual screening from an extended healthcare perspective [45].

The Australian study compared manual screening to three different AI-based screening scenarios from the healthcare system perspective [46]. All three scenarios resulted in dominant ICERs compared with manual screening, with cost savings ranging from 1% for low screening uptake to 5% for high screening uptake, with the most cost-saving scenario resulting in savings exceeding $\$400$ million (2025 USD) over 40 years.

In the Canadian study by Chakravarthy et al., from the healthcare system, the implementation of an AI-based screening approach that selects 20% of patients at highest risk of vision loss for optometrist screening resulted in savings of $\sim\$145$ million (2025 USD) for Ontario compared with downstream costs associated with inaction; the approach of screening all patients without the use of AI-software resulted in cost savings of $\sim\$61$ million (2025 USD) compared

Table 3 Key outcomes of studies that found no significant differences in visual acuity between groups with and without treatment delay

Study source and region	Details
Naravane et al. (2021), USA [33]	A ≥ 2-week delay beyond the recommended treatment period for anti-VEGF due to the COVID-19 pandemic worsened BCVA by 0.178 among patients with DME with or without PDR ($n = 34$), but this did not reach statistical significance ($p = 0.06$) CST was significantly greater at post-lockdown follow-up among these patients compared with CST pre-lockdown (341 microns versus 447 microns; $p = 0.03$)
Fassbender et al. (2016), USA [34]	No significant differences in final visual acuity were observed between patients who underwent delayed versus immediate vitrectomy for PDR, but patients in the delayed group spent significantly greater time with decreased vision preoperatively compared with patients who underwent immediate vitrectomy ($p < 0.0001$)
Bulut et al. (2024), Turkey [32]	A 6-month delay in anti-VEGF injection during the COVID-19 pandemic did not have a significant impact on the BCVA of treatment-naïve patients with DME with nonproliferative DR
Gomel et al. (2023), Israel [36]	There was no significant difference between baseline and final BCVA among patients who experienced 2–3-month treatment gaps due to the COVID-19 pandemic in 2020 (baseline: 0.45 ± 0.43 ; final: 0.47 ± 0.44 ; $p = 0.323$); no significant difference between baseline and final BCVA was seen in the 2019 cohort either, which comprised patients who did not experience treatment delays (baseline: 0.52 ± 0.44 ; final: 0.55 ± 0.41 ; $p = 0.082$)
Rush and Rush (2021), USA [35]	A longer treatment delay across patients with DME as a result of the COVID-19 pandemic was correlated with worse visual acuity prior to anti-VEGF reinitiation following the delay ($p = 0.015$), but no such correlation was observed 6 months post treatment reinitiation ($p = 0.11$)
Hentati et al. (2021) [37], USA	Across 72 patients with DME and best visual acuity of 20/25 or better at baseline, patients with early anti-VEGF treatment (within 6 months of diagnosis) experienced a loss of five ETDRS letters or more at 6 months compared with patients that experienced delayed treatment (after 6 months of diagnosis) or no treatment ($p = 0.0061$), but no such difference was observed among the groups at 12 and 24 months ($p = 0.651$ and $p = 0.447$, respectively) The authors did acknowledge that the early treatment group had more severe disease than the other groups, which could have contributed to visual acuity loss at 6 months

anti-VEGF anti-vascular endothelial growth factor, *BCVA* best-corrected visual acuity, *CST* central subfield thickness, *DME* diabetic macular edema, *ETDRS* Early Treatment Diabetic Retinopathy Study, *PDR* proliferative diabetic retinopathy, *PRP* panretinal photocoagulation, *PSM* propensity score-matched, *USA* United States

with inaction [47]. Therefore, cost savings with the AI-based approach were more than twice that of savings with the conventional approach, compared with inaction for any patient.

Two studies found that AI-based strategies for DR screening resulted in higher incremental

costs than conventional screening. One study compared organized screening (technicians, AI photo grading, optical coherence tomography, human arbitration) with 80% annual uptake to current ophthalmologist-led opportunistic screening with 2% annual uptake in China [48].

ICERs (in 2020 USD) for organized screening versus opportunistic screening over a lifetime horizon were \$2942 per QALY and \$10,259 per QALY, depending on the treatment strategy utilized for clinically significant macular edema, both of which were within the threshold of \$11,638 per QALY required to be considered highly cost-effective in China in 2020. The other study found DR diagnosis with AI to result in higher costs than DR diagnosis without AI in Brazil; both strategies resulted in similar QALY gains [49].

Lastly, a Thailand-based study by Srisubatt et al. found the AI-based screening approach to be associated with higher incremental costs (in 2025 USD) from the provider perspective (~\$78.38) and a slightly lower cost from the societal perspective (~−\$3.06) compared with human grading [50]. The authors noted that, from the provider perspective, the AI-based strategy was able to detect more referral cases, resulting in greater treatment costs. However, in the societal perspective, wherein costs incurred from bilateral blindness were considered, the AI-based screening approach resulted in cost savings.

Clinical Impact Models of AI for DR Screening

Six studies modeled the impact of AI-augmented DR screening on blindness and vision loss, five of which were cost-effectiveness analyses and one of which was a policy model. Four studies found that AI-augmented DR screening reduced blindness or vision loss compared with conventional screening techniques. A US policy model estimated the incidence of vision loss at 5 years to be reduced with the use of AI: 1535 per 100,000 in AI-screened patients, versus 1625 per 100,000 in patients screened by eye care providers. Additionally, the model estimated that 27,000 fewer Americans would experience 5-year vision loss with the implementation of AI-based screening [51].

Three studies were CEAs that also modeled the impact of AI-based DR screening on rates of blindness or vision loss [38, 46, 48]. Hu et al. found that scaling AI-based DR screening to universal coverage would prevent 38,347 and 1211 cases of blindness in non-Indigenous and Indigenous populations across Australia, respectively,

compared with current practices [46]. The study by Harding et al. estimated that the AI-based approach resulted in a 52.4–61.9% reduction in blindness in China, compared with ophthalmologist-led opportunistic screening over a lifetime horizon, depending on the type of therapy employed for macular edema [48]. Lastly, a US study by Fuller et al. estimated severe vision loss to decrease from 3.1% to 1.1% with the implementation of AI-based DR screening in the USA after 5 years [38]. In the Canadian study by Chakravarthy et al., implementation of a risk-profiling AI-based screening approach that selects 20% of patients at highest risk of vision loss for optometrist screening resulted in only a 5% difference in estimated vision loss compared with screening all unscreened patients with diabetes [47]. In particular, the non-AI approach (all unscreened patients are screened) was estimated to prevent 95% of vision loss cases, and the risk-profiling approach was estimated to prevent 90% of vision loss cases, with more than twice the amount of cost savings compared with inaction. The final cost-effectiveness analysis modeled years without blindness and found that, versus manual grading for DR, AI-based telemedicine screening may result in a very similar number of years without blindness in China [40].

Risk of Bias Assessment

Overall, 16 studies were reviewed using the Drummond and Jefferson checklist (Supplementary Tables S4 and S5), and 14 were reviewed using the Downs and Black checklist (Supplementary Tables S6 and S7). One study reported on a policy model, while two other studies were only available in abstract form and did not provide sufficient data to be included in this quality assessment.

All 16 economic evaluations assessed using the Drummond and Jefferson checklist provided adequate details regarding their research context, including, but not limited to, clear statement of the research question, economic importance, and analysis viewpoint. Reporting of outcomes was generally strong. Reporting on productivity changes was limited; most studies neither reported productivity changes separately

nor discussed their relevance (14/16). Details regarding inflation adjustments or currency conversion were absent in several cases (9/16). Other methodological features (e.g., model selection) and data handling (e.g., inclusion of supplemental analyses) were generally considered appropriate. Overall, the quality of reporting for these models was moderate to strong.

All 14 observational studies assessed using the Downs and Black checklist reported adequate details regarding their study design, statistical testing, and findings. Most studies were conducted using retrospective, noninterventional designs, and did not require blinding or randomization, nor were they required to report on adverse events and patient loss to follow-up. With most studies being conducted at the single- or multicenter level rather than at large-scale or national levels, external validity on whether patients and staff involved were representative of the entire population could not be determined for several studies. Adequate adjustment for confounding variables was completed for most studies.

DISCUSSION

This SLR sought to understand and quantify the economic, humanistic, and clinical burden associated with the lack of timely DR screening and treatment and to highlight the impact of AI tools for DR diagnosis and treatment. In general, studies encompassing this SLR found that delayed or less frequent screening of patients with diabetes was associated with overall lower screening costs but higher downstream management costs and worse visual outcomes. Further, studies that discussed the use of AI-based tools for DR screening in this SLR generally demonstrated that these tools were associated with lower costs, with several models also predicting improved patient visual outcomes.

The economic studies that evaluated variable screening frequencies found that, in general, a reduction in the frequency of screening or a delay in the timing of DR screening was not only associated with lower upfront screening costs but also with high downstream treatment costs

and QALY reductions [19, 20]. Interestingly, the study by Brodie et al. found DR screening to be less cost-effective with increasing age of diabetes diagnosis owing to the increased risk of death among older patients before they are able to benefit from treatment [21]. Nevertheless, these findings also suggest that screening earlier in younger patients with diabetes may be cost-effective and more beneficial to patients who are able to spend a longer amount of time with improved vision following treatment for DR.

Studies evaluating clinical impacts demonstrated that a delay in DR screening following diabetes diagnosis is significantly associated with worsened disease severity upon screening [26, 27]. Further, treatment delays in patients with DR, PDR, or DME are associated with either visual acuity loss, more time spent with vision loss, increased CST, or a requirement for more invasive procedures such as vitrectomy. Evidence as such underscores the importance of timely DR management.

Despite this, many patients with diabetes are unable to obtain timely screening, particularly owing to affordability and access issues. The introduction of AI-based strategies for DR screening has the potential to address these limitations. As noted by Fuller et al., drivers of savings include a reduction in unnecessary patient referrals, an increase in adherence to follow-up among patients with worsening DR, and a decrease in the incidence of severe vision loss as a result of timely treatment of vision-threatening DR [38]. Thus, although the upfront investment associated with developing and implementing AI-based algorithms may be high, eventual cost savings can be realized with lower resource needs required to maintain and sustain AI-based screening methods, as noted by Chakravarthy et al. [47]. Additionally, as demonstrated by Hu et al., although there is an increase in direct medical costs, including costs associated with consultation, angiography, optical coherence tomography, anti-VEGF treatment, and vitrectomy, as a result of the scalability of AI-based screening to capture more DR cases, costs are eventually offset by reductions in screening costs and costs associated with care for blindness [46]. It should be noted that the cost-effectiveness of

AI-based technologies is often contingent on the specificity of algorithms, defined as their ability to detect true negatives. Unnecessary referrals and diagnostics due to excessive false positives may mitigate the cost savings associated with AI-based screening. This necessitates the sensitivity and specificity of AI algorithms to be tested across diverse and distinct populations prior to being introduced into clinical practice.

The majority of studies in this SLR predicted an overall reduction in DR-related vision loss or blindness with the introduction of AI-augmented screening. Channa et al. noted that the effectiveness of AI-based algorithms is likely driven by the point-of-care availability of AI-based DR screening tools and immediate diagnostic outputs, which motivate patients to accept screening and referral [51]. This is especially important in rural, geographically isolated, or marginalized communities where access to screening and skilled healthcare workers is limited.

The strength of this study is supported through the utilization of a robust SLR approach and directed supplementary searching to comprehensively address the research questions. The approach adhered to best practices defined by the Cochrane Collaboration. Multiple quality assessment methods were employed to ensure that the identified study designs were appropriately appraised. Overall, studies were generally deemed to be of good quality. This SLR has some limitations. In general, there was a lack of studies discussing the humanistic impact of DR diagnosis and treatment timing on patients, although it is well-established that gradual vision impairment as a result of DR can affect several quality of life domains. Additionally, there was a lack of US-specific data. Studies, specifically the economic analyses, were based on geographically heterogeneous populations, spanning drastically different healthcare systems, making it uncertain whether the cost-effectiveness outcomes and modeled benefits of AI translate consistently across countries and their populations and health systems. Additionally, other aspects, including the specificity/sensitivity of the AI algorithm evaluated, number of cases screened,

and purpose of the algorithm (e.g., for diagnosis, referral, or grading), may have limited the comparability and generalizability of findings. Further, there was a lack of studies describing the real-world impact of AI-based approaches on patient outcomes. Given the novelty of AI-based DR screening, more real-world data on patient outcomes are anticipated in the future. Lastly, for studies that did not report cost year, it was assumed that cost year was the same as publication year. Given that many studies utilize data from 1 to 3 years prior, and that healthcare cost inflation diverges from general consumer price indices, reported costs might differ slightly from actual costs. However, any cost trends and overall conclusion still apply in these cases.

Overall, this review has highlighted the economic and patient-related consequences of delayed DR screening and treatment. AI-based DR screening tools can increase the affordability and accessibility of screening for underserved populations, which has the potential to prevent vision loss and blindness associated with DR. Future real-world studies evaluating AI-based diabetic retinopathy screening across more diverse healthcare systems are required to confirm whether the modeled clinical benefits translate into improved outcomes for patients.

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Declarations

Conflict of Interest. Dr. Firas Rahhal is a consultant for Boehringer Ingelheim. Jun Zhang and Munia Mukherjee are employees of Boehringer Ingelheim Pharmaceuticals, Inc.

Ethical Approval. This article is based on previously conducted studies and does not contain any new studies with human participants or animals performed by any of the authors; therefore, ethical approval was not required.

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